MATH 110

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Contents

1 First Order PDEs

1.1 Lecture 1 - Introduction to PDEs

What is a ODE? Recall that an ordinary differential equation (ODE) are equations for functions of a single variable, $u(t),\ t \in \mathbb{R}$

$$\frac{du}{dt} = u(1-u) \text{ is an ODE}$$

Here we call t the independent variable and u the state/dependent variable. In general, an ODE has the form $F(t) = (t, u, \frac{du}{dt}, \frac{d^2u}{dt^2}, \dots)$

Note that the state variable can have many components.

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \in \mathbb{R}^2$$

then

$$\frac{du}{dt} = \mathbf{u} \text{ is an ODE}$$

What is a PDE? A PDE is an equation for a function with more than 1 independent variable, e.g. $u(x, y)$.

$$\begin{aligned} \text{Notation: } u_x &= \frac{\partial u}{\partial x} = \partial_x u \\ u_y &= \frac{\partial u}{\partial y} = \partial_y u \\ &\text{etc.} \end{aligned}$$

Examples of PDEs

- Heat/diffusion equation for $u(x, t)$

$$u_t = k u_{xx} \quad k \text{ constant}$$

- Wave equation for $u(x, t)$

$$u_{tt} = c^2 u_{xx} \quad c \text{ constant}$$

- Laplace's equation for $u(x,y)$

$$u_{xx} + u_{yy} = 0$$

$\boxed{\text{ex}}$ Consider the heat flow in a bar of length l

- $u(x,t)$ is the temperature of the bar at position x, time t
- Suppose the ends are held at temperature 0, and the initial temperature is $\phi(x)$
- The temperature evolves according to the heat equation $u_t = ku_{xx}$ where $t \geq 0, 0 \leq x \leq l$, k constant.

We also have:

- Boundary conditions (BCs):

$$u(0,t) = u(l,t) = 0 \quad \forall t$$

- Initial conditions (ICs):

$$u(x,0) = \phi(x)$$

<u>Remarks</u>

- The order of the PDE is the order of the highest derivative that occurs.
- The heat equation above is 2^{nd} order
- A general 2^{nd} order PDE for $u(x,t)$ is:

$$G(x,t,u,u_x,u_t,u_{xx},u_{tt},u_{xt}) = 0$$

- A PDE in the above form is <u>linear</u> if G is a linear function of u and all of its derivatives. It is nonlinear otherwise.
- A linear PDE is <u>homogeneous</u> if every term contains u or some derivative of u. It is nonhomogeneous/inhomogeneous otherwise.

$\boxed{\text{ex}}$ Consider the linearity and homogeneity of the following equations.

1. $u_t + e^{x^2 t} u_{xt} = 0$
   is a linear and homogeneous PDE.

2. $u_t + 3x u_{xx} = \frac{t}{x}$
   is a linear and inhomogeneous PDE because of the $\frac{t}{x}$ term.

3. $u_t + u u_x = 0$
   is a nonlinear PDE because of the $u u_x$ term.

4. $u_{tt} + u_x + sin(u) = 0$
   is nonlinear because of the $sin(u)$ term.

Operating notation & linearity

We can write the heat equation $u_t - k u_{xx} = 0$ as $Lu = 0$ where

$$L = (\partial_t - k\partial_x^2)$$

We call L the differential operator. L acts on (smooth) functions $u(x,t)$ to produce a new function.

$$\begin{aligned} L(u) = Lu &= (\delta_t - k\partial_x^2)u \\ &= u_t - k u_{xx} \end{aligned}$$

L is a linear operator iff

$$\begin{aligned} L(u+v) &= Lu + Lv \\ &\forall \text{ functions } u, v \end{aligned}$$

and

$$\begin{aligned} L(cu) &= cLu \\ &\forall \text{ constants } c \in \mathbb{R} \\ &\forall \text{ functions } u \end{aligned}$$

$\boxed{\text{ex}}$ Check if the heat equation has a linear differential operator.

1.

$$\begin{aligned} L(u+v) &= (\partial_t + k\partial_x^2)(u+v) \\ &= u_t + k u_{xx} + v_t + k v_{xx} \\ &= Lu + Lv \quad \checkmark \end{aligned}$$

2.

$$\begin{aligned} L(cu) &= cu_t + cku_{xx} \\ &= cLu \text{ for constant } c \quad \checkmark \end{aligned}$$

Remarks

- If L is a linear operator then $Lu = f$ is a linear PDE

$$\underbrace{u_t + ku_{xx}}_{Lu} = \underbrace{e^{x^2 t}}_{f(x,t)} \text{ is linear}$$

- If L is a linear operator then the PDE $Lu = f$ is
  - homogeneous if $f = 0$
  - inhomogeneous if $f \neq 0$
- A linear, homogeneous PDE $Lu = 0$ has the superposition principle
- If u_1, u_2 are solutions to the PDE then $au_1 + bu_2$ is a solution for constants a, b.

$$\begin{aligned} Lu_1 &= Lu_2 = 0 \\ \Rightarrow L(au_1 + bu_2) &= 0 \end{aligned}$$

- The distinction between linear & nonlinear PDEs is very important.

Solutions to PDEs

- A solution to a PDE is a function u in which satisfies the PDE and any BCs/ICs upon direct substitution.
- The general solution to an ODE involves arbitrary constants. The general solution to a PDE involves arbitrary functions.
- Typically, we are interested in specific solutions.

$\boxed{\text{ex}}$ Consider the heat equation $u_t - u_{xx} = 0$.

$$\begin{aligned} u_1(x,t) &= x^2 + 2t \text{ is a soln} \\ \to \partial_t u_1 - \partial_x^2 u_1 &= 2 - 2 = 0 \\ u_2(x,t) &= e^{-t} \sin x \text{ is a soln} \\ \to \partial_t u_2 - \partial_x^2 u_2 &= -e^{-t}\sin x + e^{-t}\sin x = 0 \end{aligned}$$

$\boxed{\text{ex}}$ Consider the equation $u_x = tx$ (1st order, linear, inhomogeneous)

$$\begin{aligned} Lu = f \text{ with } L &= \partial_x \\ f(x,t) &= tx \end{aligned}$$

We can solve this by direct integration on both sides wrt x.

$$u(x,t) = \frac{1}{2}tx^2 + A(t)$$

Where $A(t)$ is an arbitrary function of t

$\boxed{\text{ex}}$ Consider the equation $u_{tt} - 4u = 0$ (2nd order, linear, homogeneous)

$$\begin{aligned} Lu &= f \\ \to L &= (\partial_t^2 - 4) \\ f(x,t) &= 0 \end{aligned}$$

There is no explicit x-dependence

- We can treat this as an ODE with x as a parameter.
- 'Constants' depend on x
- General solution:

$$u(x,t) = A(x)e^{-2t} + B(x)e^{2t}$$

1.2 Lecture 2 - Characteristics

Recap:

- Heat equation for $u(x,t)$

$$u_t - ku_{xx} = 0, \quad k > 0, \text{constant}$$

- We can write the heat equation as $Lu = 0$, where $L = (\partial_t - k\partial_x^2)$ is a differential operator.
- L is linear $\leftrightarrow$ $L(u+v) = Lu + Lv$ and $L(cu) = cLu$ $\forall$ functions u, v, constant c
- PDE: $Lu = f$ is linear if L is linear
- Linear PDE: $Lu = f$ is homogeneous if $f = 0$, inhomogeneous otherwise.

First-order linear PDE

Consider functions $u(x, y)$

$$u : \mathbb{R}^2 \to \mathbb{R}$$

Often we think of x, y as spatial variable and t as time variables. The simplest PDE we can have following this is:

$$u_x = 0$$

Which is 1st-order, linear, homogeneous. Also note that there is no explicit y-dependence. Therefore we can treat this like an ODE, and solve with direct integration of both sides.

$$u(x, y) = f(y)$$

where $f(y)$ is an arbitrary function of y.

Now consider $au_x + bu_y = 0$, which is a 1st-order, linear, homogeneous PDE with constants $a, b \in \mathbb{R}$.

$$\begin{aligned} \text{Let } V &= (a, b) \in \mathbb{R}^2 \\ \text{Recall } \nabla u &= (u_x, u_y) \\ \text{Then } V \cdot \nabla u &= (a, b) \cdot (u_x, u_y) \\ &= au_x + bu_y = 0 \end{aligned}$$

$\Rightarrow$ Directional derivative in direction V is 0.
$\Rightarrow$ u is constant on lines parallel to V.

Let $V^\perp = (b, -a)$, so that $V \cdot V^\perp = 0$.
Lines parallel to V satisfy

$$\begin{aligned} &V^\perp \cdot (x, y) = \text{constant} \\ &\Rightarrow bx - ay = \text{constant} \end{aligned}$$

Then u is constant on lines parallel to V.

$$\Rightarrow u(x, y) = f(bx - ay)$$
$$\text{for an arbitrary function of } f$$

<u>Check</u>:

$$\begin{aligned} &(a\partial_x + b\partial_y)\ f(bx - ay) \\ &= abf' - baf' = 0 \\ &\Rightarrow f' \text{ is a derivative of } f : \mathbb{R} \to \mathbb{R} \end{aligned}$$

Characteristic Coordinates. In this example, the lines parallel to V are called characteristics. Here u is constant along the characteristics.

We can write points (x, y) in characteristic coordinates (ξ, η). Where ξ is the x-intercept of the characteristic line, and η is the distance along that line to the point (x, y).

$$\begin{aligned} \text{So } x &= \xi + a\eta = x(\xi, \eta) \\ y &= b\eta = y(\xi, \eta) \\ \xi &= x - a\eta \\ \eta &= \frac{y}{b} \\ \xi &= x - \frac{a}{b}y \\ b\xi &= bx - ay \end{aligned}$$

Transform PDE to (ξ, η) coordinates.

Chain rule:

$$\frac{\partial u}{\partial \eta} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \eta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \eta}$$
$$\Rightarrow u_\eta = au_x + bu_y = 0$$

Using characteristic coordinates (ξ, η), we've transformed to the PDE $u_\eta = 0$ for $u(\xi, \eta)$.

Solution:

$$u = f(\xi)$$
$$\text{where } \xi = \frac{1}{b}(bx - ay)$$
$$\Rightarrow u = g(bx - ay)$$

Where g is an arbitrary function.

Remarks:

- η is coordinates along characteristics
- u is constant on characteristics $\Rightarrow u_\eta = 0$
- ξ labels different characteristics. There are other choices for ξ but any choice $\tilde{\xi}$ would have $\tilde{\xi} = h(\xi)$

Method of Characteristics
Method for solving PDEs of form

$$a(x, y)u_x + b(x, y)u_y = 0 \quad \text{for } u(x, y)$$

Idea: Let $A(x, y) = (a(x, y), b(x, y))$
The directional derivative $A \cdot \nabla u = 0$
$\Rightarrow u$ is constant along curves w/ slope A.

- These are characteristic curves.
- Find coord η along chars
- Find coord ξ which labels chars
- PDE transforms to $u_\eta = 0$, general soln: $u = f(\xi)$

$\boxed{\text{ex}}$ $u_x + yu_y = 0$
The directional derivative along $(1, y)$ are 0.
The chars $(x(\eta), y(\eta))$ satisfy

$$\frac{dx}{d\eta} = 1, \ \frac{dy}{d\eta} = y$$

At $\eta = 0$, take $(x(0), y(0)) = (0, \xi)$
Solve:

$$\begin{aligned}
&\frac{dx}{d\eta} = 1,\ x(0) = 0 \\
&\Rightarrow x = \eta \\
&\frac{dy}{d\eta} = y,\ y(0) = \xi \\
&\Rightarrow y = \xi e^{\eta}
\end{aligned}$$

This follows that $\xi = ye^{-x}$.

$$\begin{aligned}
\Rightarrow u_\eta &= u_x \frac{\partial x}{\partial \eta} + u_y \frac{\partial y}{\partial \eta} \\
&= u_x + \xi e^{\eta} u_y \\
&= u_x + y u_y = 0 \\
\Rightarrow u_\eta &= 0 \\
\Rightarrow u &= f(\xi) = f(ye^{-x}) \text{ with arbitrary function } f
\end{aligned}$$

<u>General solution</u>:
Suppose we are given $u(0, y) = y^2$

$$\begin{aligned}
&\Rightarrow \text{Specific solution} \\
&u(0, y) = f(y) = y^2 \\
&\Rightarrow u(x, y) = (ye^{-x})^2
\end{aligned}$$

Alternatively, from parametric equations,

$$\begin{aligned}
&\frac{dx}{d\eta} = 1 \qquad \frac{dy}{d\eta} = y \\
&\Rightarrow \frac{dy}{dx} = y \\
&\Rightarrow y = Ce^x
\end{aligned}$$

$\Rightarrow u$ is constant along chars.
$\Rightarrow u(x, Ce^x) = u(0, C)$
$= u(0, ye^{-x})$

$= f(e^{-x}y)$ for arbitrary f

$\boxed{\text{ex}}$ $u_x + 2xy^2u_y = 0$
Characteristic curves: $(x(\eta), y(\eta))$

$$\frac{dx}{d\eta} = 1 \qquad \frac{dy}{d\eta} = 2xy^2$$
$$\frac{dy}{dx} = 2xy^2$$

$$\int \frac{1}{y^2}dy = \int 2x dx$$
$$-\frac{1}{y} = x^2 - c$$
$$y = \frac{1}{c - x^2}$$

u is constant along chars.
$\Rightarrow u(x,y) = f(c) = f(x^2 + \frac{1}{y})$ is the general soln.

<u>Advection, transport</u>
Consider $u(x,t)$ describing quantity u at position x, time t.

$$u_t + cu_x = 0 \quad (c > 0)$$

Chars: $(t(\eta), x(\eta))$ satisfy

$$\frac{dt}{d\eta} = 1 \qquad \frac{dx}{d\eta} = c$$
$$\Rightarrow \frac{dx}{dt} = c$$
$$\Rightarrow x = ct + \xi \quad \text{via integration}$$

General solution:

$$u = f(\xi) = f(x - ct)$$

1.3 Lecture 3 - Conservation Laws

Recap:

- Solve PDEs of the form $a(x,y)u_x + b(x,y)u_y = 0$
- The char curves $(x(\eta), y(\eta))$ satisfy

$$\frac{dx}{d\eta} = a(x,y) \qquad \frac{dy}{d\eta} = b(x,y)$$

- u is constant along chars.

$\boxed{\text{ex}}$ Given $\frac{1}{x^2}u_x + u_y = 0$

$$\frac{dx}{d\eta} = \frac{1}{x^2} \qquad \frac{dy}{d\eta} = 1$$

Method 1: divide to get parametric eq.

$$\frac{dy}{dx} = x^2$$
$$y = \frac{1}{3}x^3 + \xi$$

where constant ξ labels the char.

The general solution is $u = f(\xi) = f(y - \frac{1}{3}x^3)$ for an arbitrary function f. This can be confirmed by plugging u back into the PDE.

Method 2: solve parametric equations.

$$\frac{dx}{d\eta} = \frac{1}{x^2} \qquad \frac{dy}{d\eta} = 1$$
$$\Rightarrow \int x^2 dx = d\eta \qquad y = \eta + B$$
$$\Rightarrow \frac{1}{3}x^3 = \eta + A \qquad y = \eta + B$$

We can absorb the two constants into one ξ

$$\Rightarrow \frac{1}{3}x^3 = \eta + \xi \qquad y = \eta$$
$$\Rightarrow y = \frac{1}{3}x^3 - \xi$$

We should check that the chars cover the whole plane, i.e. no chars have been 'missed', via sketching or finding

$$\eta = \eta(x, y)$$
$$\xi = \xi(x, y)$$

Coord transformation is also useful

$$\begin{matrix} \frac{1}{3}x^3 = \eta + \xi \\ y = \eta \end{matrix} \Rightarrow \begin{matrix} x = (3\eta + 3\xi)^{\frac{1}{3}} \\ y = \eta \end{matrix}$$

This transform our PDE

$$\frac{1}{x^2}u_x + u_y = 0 \Leftrightarrow u_\eta = 0$$
$$\text{Gen soln: } u = f(\xi)$$
$$= f(\frac{1}{3}x^3 - y)$$

with arbitrary function f

$\boxed{\text{ex}}$ Advection

Consider function $u(x, t)$ describing a quantity u at position x and time t

$$u_t + cu_x = 0 \qquad \text{(Advection)}$$

with constant $c > 0$

Chars $(t(\eta), x(\eta))$ satisfy

$$\frac{dt}{d\eta} = 1 \qquad \frac{dx}{d\eta} = c$$

Solve

$$t = \eta + A$$
$$x = c\eta + B$$

Absorb into one constant using $A = 0, B = \xi$

$$\Rightarrow t = \eta$$
$$x = c\eta + \xi$$
$$\Rightarrow x = ct + \xi$$

By inverting, $\eta = t,\ \xi = x - ct$.
Now we have an transformed PDE

$$u_\eta = u_t + cu_x = 0$$

General solution is

$$\begin{aligned} u(x,t) &= f(\xi) \\ &= f(x - ct) \end{aligned}$$

for an arbitrary f.

Suppose we had initial condition $u(x, o) = \phi(x)$, then $u(x, 0) = f(x) = \phi(x) \Rightarrow$ solution is $u(x,t) = \phi(a - ct)$.

<u>Remarks</u>:
We can solve more complicated PDEs using chars, for example,

$$u_t + cu_x = f(u)$$

Transform to coords (ξ, η) with $t = \eta$, $x = c\eta + \xi$

$$u\eta = f(u)$$

which can be solved as an ODE

Conservation Laws. (Logan 1.2)

- Where do PDEs come from?
- For a conservated quantity:

$$\begin{array}{c} \text{rate of change} \\ \text{in domain} \end{array} \implies \begin{array}{c} \text{net flow into domain} \\ +\text{ creation in domain} \end{array}$$

e.g. Animal population in a fixed region

<u>Quantitatively</u>

- Let $u(x,t)$ be density of some quantity in 1d domain.
- The amount of quantity in section dx is $u(x,t)Adx$.

- Let $\phi(x,t)$ be the flux, which is the rate of quantity crossing section x at time t. Note that $\phi(x,t)dt$ is the amount of stuff, and so is $u(x,t)dx$.
- $A\phi(x,t)$ is amount of quantity passing x at time t
- Let $f(x,t)$ be the rate at which quantity is created.
- $f(x,t)Adx$ is amount of quantity created in width dx per unit time.

Remarks:

- $\phi(x,t) > 0 \implies$ flux is L to R
- $\phi(x,t) < 0 \implies$ flux is R to L
- $f(x,t) > 0$ is source at x (creation)
- $f(x,t) < 0$ is sink at x (destroyed)

Balance:

$$\begin{array}{c}\text{change in}\\ \text{quantity in}\\ \text{section}[a,b]\end{array} = \text{flux} + \text{creation}$$

$$\begin{aligned}\Rightarrow \frac{d}{dt}\int_a^b u(x,t)Adx &= A\phi(a,t) - A\phi(b,t) + \int_a^b f(x,t)Adx\\ \Rightarrow \int_a^b u_t Adx &= -\int_a^b \phi_x Adx + \int_a^b fAdx\\ \Rightarrow \int_a^b (u_t + \phi_x - f)dx &= 0\end{aligned}$$

This is true for any domain $[a,b]$

$$u_t + \phi = f \text{ where}$$

$$\begin{array}{c}\text{change in}\\ \text{quantity}\end{array} = \text{net flux} + \text{net creation}$$

Remarks:
Can get a PDE for u by

- Specifying $f(x,t)$
- Choosing a relationship between ϕ and u, u_x, etc. This is called constitutive law.
- E.g. $\phi = cu \to$ advection.

<u>Diffusion</u>:

- Conservation law with no sources

$$u_t + \phi_x = 0$$

- Suppose $u(x,t)$ is density of a gas of particles. Then
  - Particles move from high u to low u.
  - Flux increase with density gradient.
  - Fick's law:

    $$\phi(x,t) = -Du_x$$

    where $D > 0$ is diffusion constant.
  - This gives the diffusion equation

    $$u_t = Du_{xx}$$

    which is the same as the heat equation.

<u>Remark</u>:

- Can combine diffusion with other effects.
  e.g. Advection-diffusion eq.

  $$\text{flux: } \phi = cu - Du_x$$

  <u>Balance</u>:

  $$\begin{aligned} u_t + \phi_x &= 0 \\ u_t + cu_x &= Du_{xx} \end{aligned}$$

2 Intro to Heat, Wave, Laplace's equations

2.1 Lecture 4 - Diffusion Equation

Recap:

- Conservation Laws (Logan 1.2)
- Density: $u(x,t)$
- Flux: $\phi(x,t)$ (movement of u)
- Sources: $f(x,t)$ (creation of u)
- Balance in domain $[a,b]$

$$\frac{d}{dt}\int_a^b Au(x,t)dx = A\phi(a,t) - A\phi(b,t) + \int_a^b Af(x,t)dx$$

$$\int_a^b u_t dx = -\int_a^b \phi_x dx + \int_a^b f dx$$

$$\int_a^b (u_t + \phi_x - f)dx = 0$$

This holds for any $[a,b]$

$$u_t + \phi_x = f$$

$$\begin{array}{c}\text{change in}\\ \text{quantity}\end{array} + \text{flux} = \text{source}$$

Diffusion equation.
Suppose $u(x,t)$ is density/concentration of substance. We have Fick's Law for flux:

$$\phi(x,t) = -Du_x,\ D > 0$$

- Movement from high u to low u
- Steeper gradient gives higher flux

Conservation/balance Law:

$$\begin{aligned} u_t + \phi_x &= 0 \\ \Rightarrow u_t &= D u_{xx} \qquad \text{(Diffusion/heat eq)} \end{aligned}$$

Remarks:

- D is called diffusivity or thermal conductivity.
- Some physical settings have a non-constant diffusivity $D(x)$

  $$\phi = -D(x) u_x$$

  This yields the PDE

  $$\begin{aligned} u_t &= (D u_x)_x \\ &= D' u_x + D u_{xx} \\ \text{where } D' &= \frac{dD}{dx} \end{aligned}$$

- Can also find nonlinear diffusion equations where $D = D(u)$

  $$u_t = (D(u) u_x)_x$$

- We can combine diffusion with other terms, e.g. diffusion-advection

  $$\begin{aligned} u_t + c u_x &= D u_{xx} \\ \phi(x,t) &= -D u_x + c u \end{aligned}$$

Dimensions/units:
Suppose $u(x,t)$ is temperature, then

$$\begin{aligned} [u] &= \text{temp} \\ [u_t] &= \frac{\text{temp}}{\text{time}} \\ [u_{xx}] &= \frac{\text{temp}}{\text{length}^2} \end{aligned}$$

So $u_t = D u_{xx}$

$$\Rightarrow [D] = \frac{\text{length}^2}{\text{time}}$$

Consider heat flow in a region of length L. The time-scale T for temperature change is

$$T = \frac{L^2}{D} \text{ so } D = \frac{L^2}{T}$$

Boundary & Initial Conditions. Diffusion eqs usually have BCs and ICs. Consider heat flow in 1d with domain $0 < x < l$.

IC has form

$$u(x, 0) = u_0(x)$$

where $u_0(x)$ is initial temperature profile.

There are 3 types of common BCs.

1. Dirichlet BCs
   Specify u at the boundary.
   $$u(0, t) = g_1(t)$$
   $$u(l, t) = g_2(t)$$
   where g_1, g_2 are given i.e. u is known at boundary.
2. Neumann BCs
   Specify u_x at the boundary.
   $$u_x(0, t) = h_1(t)$$
   $$u_x(l, t) = h_2(t)$$
   where h_1, h_2 are given, i.e. flux is known at boundary.
3. Robin BCs
   Specify u, u_x given at boundary.
   $$au(0, t) + bu_x(0, t) = \psi_1(t)$$
   $$au(l, t) + bu_x(l, t) = \psi_2(t)$$
   where ψ_1, ψ_2 are given. E.g. Newton's Law of Cooling states that heat flow $\propto$ temp difference between object and environment.
   $$bu_x(0, t) = -au(0, t) + \psi_1(t)$$
   $$\text{flux} = \text{object} + \text{environment}$$

Remarks:

- A well-posed problem is a PDE with ICs and BCs s.t. it has a unique solution with a correct physical meaning (stability).
- The above terminology for BCs applies to general PDEs.

Steady-state solutions. Many time-dependent PDEs have solutions which approach steady states $u = u(x)$, which are time-independent. E.g. diffusion/heat equation:

$$u_t = Du_{xx}$$

Steady state solution $u(x)$ has $u_t = 0$, so

$$Du'' = 0 \quad (u' = u_x)$$

This gives a 2nd-order ODE, where

$$u(x) = Ax + B$$

for constants A, B set by BCs.

$\boxed{\text{ex}}$ Consider the diffusion-decay equation

$$\begin{gathered} u_t = Du_{xx} - ru \\ 0 < x < L,\ t, D, r > 0 \end{gathered}$$

u has the flux $\phi = -Du_x$ and is destroyed at rate ru. Take BCs:

$$\begin{gathered} u(0,t) = 0 \\ -Du_x(L,t) = -1 \\ t > 0 \end{gathered}$$

and IC

$$\begin{gathered} u(x,0) = g(x) \\ 0 < x < L \end{gathered}$$

Remarks:
This model could describe a diffusing pollutant (Du_{xx}) which decays at rate ru.

$$\begin{aligned}\text{At } 0: & \; u(0,t) = 0 \\ & \Rightarrow u \text{ is held at } 0 \\ \text{At } L: & -Du_x(L,t) = -1 \\ & \Rightarrow \text{constant flux of } u \text{ into domain}\end{aligned}$$

Steady state problem:

$$u = u(x), \text{ so } u_t = 0$$

BVP:

$$\begin{aligned}Du'' - ru &= 0 \\ 0 < x &< L\end{aligned}$$

BCs: $u(0) = 0$, $u'(L) = \frac{1}{D}$
Here we ignore IC and assume transients have decayed.
Solution of BVP:

$$\begin{aligned}u'' &= \frac{r}{D}u \\ u(0) = 0,\ u'(L) &= \frac{1}{D}\end{aligned}$$

General solution:

$$u = A\sinh\left(x\sqrt{\frac{r}{D}}\right) + B\cosh\left(x\sqrt{\frac{r}{D}}\right) \quad \text{consts } A, B$$

BC1:

$$\begin{aligned}& u(0) = 0 \\ \Rightarrow\ & B = 0 \\ \Rightarrow\ & u(x) = A\sinh\left(x\sqrt{\frac{r}{D}}\right)\end{aligned}$$

BC2:

$$\begin{aligned} u'(L) &= \frac{1}{D} \\ \Rightarrow A\sqrt{\frac{r}{D}}\cosh\left(L\sqrt{\frac{r}{D}}\right) &= \frac{1}{D} \\ \Rightarrow u(x) &= \frac{\sinh\left(x\sqrt{\frac{r}{D}}\right)}{\sqrt{rD}\cosh\left(L\sqrt{\frac{r}{D}}\right)} \end{aligned}$$

Remarks:

- An ODE BVP may not have a solution. Physically the PDE may not have a steady state solution.
- The steady state solution could be unstable in general. (Here it turns out that $\lim_{t\to\infty} u(x,t) \to$ steady state.)

Summary. Diffusion in 1d, for $u(x,t)$

$$\begin{aligned} u_t &= Du_{xx} \\ D &> 0 \end{aligned}$$

Can consider higher dimensions for $u(x,y,t)$

Preview: Vibrations and waves
Consider the 1d wave equation for $u(x,t)$:

$$u_{tt} = C^2 u_{xx}$$

This describes, for example, waves on a string. C has dimensions length/time and is the speed of waves. 2nd-order in time $\to$ waves (contrasts with 1st-order in time $\to$ diffusion).

We have the string fixed at length $x = 0, L$ and $u(x,t)$ is the height of the string at time t. And we have the following equation of motion for the string

$$u_{tt} = \sqrt{\frac{T}{\rho}} u_{xx}$$

where T is string tension, and ρ is the density of string.

2.2 Lecture 5 - Wave Equation

Recap:

- Diffusion: $u_t = Du_{xx}$, $D > 0$
- Wave: $u_{tt} = c^2 u_{xx}$, c is speed

Vibrations of a string.

- String fixed at $x = 0, x = l$.
- $u(x,t)$ is the height of the string at time t, position x
- Density $\rho = \frac{\text{mass}}{\text{length}}$

Assumptions:

- Small deflections, i.e. $|u_x|$ is small.
- Only force acting on string is tension (no gravity, damping, etc.)
- No horizontal motion of string (only vertical)

Let $T(x,t)$ be the tension, T acts tangent to the string.
Let $\theta(x,t)$ be the angle of the tangent to the string, so

$$\cos\theta = \frac{1}{\sqrt{1+u_x^2}} \qquad \sin\theta = \frac{u_x}{\sqrt{1+u_x^2}}$$

So T is the tension force along the string. Therefore, total force on the string in region $[a,b]$ is

$$\text{horiz:} \quad T\cos\theta\Big|_a^b = \frac{T}{\sqrt{1+u_x^2}}\Bigg|_a^b$$

$$\text{vert:} \quad T\sin\theta\Big|_a^b = \frac{Tu_x}{\sqrt{1+u_x^2}}\Bigg|_a^b$$

We can simplify this using the small deflection assumption, $|u_x| \ll 1$

$$\begin{aligned}\text{Taylor series} : (1+u_x^2)^{\frac{1}{2}} &= 1 + \frac{1}{2}u_x^2 - \frac{1}{8}u_x^4 + \dots \\ &\approx 1\end{aligned}$$

where each u_x term gets smaller.

So the forces become

$$\text{horiz}: \quad T\cos\theta\Big|_a^b \approx T\Big|_a^b = \int_a^b T_x dx$$
$$\text{vert}: \quad T\sin\theta\Big|_a^b \approx Tu_x\Big|_a^b = \int_a^b (Tu_x)_x dx$$

The accelerations are

$$\text{horiz}: \quad 0 \quad \text{(by assumption)}$$
$$\text{vert}: \quad \int_a^b u_{tt} dx$$

Using Newton's 2nd Law, $F = ma$

$$\text{horiz}: \quad \int_a^b T_x dx = 0$$
$$\text{vert}: \quad \int_a^b (Tu_x)_x dx = \int_a^b \rho u_{tt} dx$$

This holds for all intervals $[a, b]$
$\Rightarrow T_x = 0$ so $T(x,t) = T(t)$
and $(Tu_x)_x = Tu_{xx}$

$$\text{vert}: Tu_{xx} = \rho u_{tt}$$
$$\Rightarrow u_{tt} = c^2 u_{xx} \text{ for } c = \sqrt{\frac{T}{\rho}}$$

<u>Remarks</u>:

- Typically assume $T(t)$ is also time-independent so $T = \text{const}$, and speed $c = \sqrt{\frac{T}{\rho}}$ is const.
- For constant c, the general solution is

  $$u(x,t) = F(x - ct) + G(x + ct)$$

  where F is a right-moving wave, and G is a left-moving wave

- For a plucked string, our BCs are

$$u(0,t) = u(l,t) = 0$$

- To get a specific solution, we need <u>2 ICs</u>, since the equation is 2nd order in time. E.g. we need

$$\begin{aligned} \text{Init position} : u(x,0) &= f(x) \\ \text{Init velocity} : u(x,0) &= g(x) \\ \text{for } 0 < x < l \end{aligned}$$

- There are many variations, e.g.:
  - Air resistance term, ru_t:

$$u_{tt} - c^2 u_{xx} + ru_t = 0$$

  - External force, $f(x,t)$:

$$u_{tt} - c^2 u_{xx} = f(x,t)$$

Standing waves. Consider a vibrating string

$$\begin{aligned} u_{tt} &= c^2 u_{xx} \\ \text{BCs} : u(0,t) &= u(l,t) = 0 \end{aligned}$$

Standing waves are solutions of the form

$$\begin{aligned} u(x,t) =& \phi(x) \cos \omega t \\ &\text{or } \phi(x) \sin \omega t \\ &\text{or in general } \phi(x) e^{i\omega t} \end{aligned}$$

Each part of the string oscillates with:

- Same freq ω, in phase.
- Different amplitude $\phi(x)$

Solutions only exist for certain ω

$$\begin{gathered}\text{substitute } u = \phi(x)\cos\omega t \text{ into wave eq} \\ -\omega^2\phi\cos\omega t = c^2\phi''\cos\omega t \\ \text{with } \phi(0)\cos\omega t = \phi(l)\cos\omega t = 0\ \forall\ t\end{gathered}$$

This gives us an ODE BVP:

$$\phi'' = -\frac{\omega^2}{c^2}\phi,\ \phi(0) = \phi(l) = 0$$

We have the following general solution

$$\phi(x) = A\cos\frac{\omega x}{c} + B\sin\frac{\omega x}{c}$$

BC1: $\phi(0) = A = 0$

$$\phi(x) = B\sin\frac{\omega x}{c}$$

BC2: $\phi(l) = B\sin\frac{\omega l}{c}$

$$B = 0 \text{ or } \sin\frac{\omega l}{c} = 0$$

If $B = 0$ then $\phi(x) = 0$ and $u(x,t) = 0$. We are interested in nonzero solutions so we need $\sin\frac{\omega l}{c} = 0$

<u>Note</u>: l and c are fixed by the problem, ω is the frequency of the standing wave. The condition $\sin\frac{\omega l}{c} = 0$ means the allowed frequencies ω of a standing wave satisfy $\sin\frac{\omega l}{c} = 0$

$$\Rightarrow \frac{\omega l}{c} = n\pi \text{ for } n \in \mathbb{Z}$$

Allowed frequencies: $\omega_n = \frac{n\pi c}{l}$
Modes: $\phi_n(x) = \sin\omega_n x = \sin\left(\frac{n\pi x}{l}\right),\ n \in \mathbb{Z}$

- $B\phi_n(x)$ satisfies the BVP for any constant B, modes defined up to a multiplicative constant.
- $n = 0$ gives $\phi_n = 0$ so it is trivial.

- $n < 0$ gives the same mode as $n > 0$ up to a sign: $\phi_n(x) = -\phi_{-n}(x)$
- We can take the modes to have $n > 0$

$$\phi_n(x) = \sin\left(\frac{n\pi x}{l}\right),\ n = 1, 2, 3, \ldots$$

The standing waves are

$$\begin{aligned} u_n(x,t) &= \cos\left(\frac{n\pi ct}{l}\right)\sin\left(\frac{n\pi x}{l}\right) \\ &= \cos(\omega_n t)\phi_n(x),\ n = 1, 2, 3, \ldots \end{aligned}$$

Remarks:

- The BVP problem is an eigenvalue problem. The frequencies ω_n are the c-values and modes ϕ_n are eigenfunctions.
- Eigenfunctions ϕ_n are defined up to a constant multiplier, like eigenvectors.
- The frequencies ω_n are the fundamental frequencies of the string.

Quantum Mechanics (Logan 1.6)
Consider a particle in 1d. Classically, the particle has a position $x \in \mathbb{R}$. In quantum mechanics, the state of the particle is a function $\Psi(x,t) \in \mathbb{C}$ called the wave function. The probability density of finding the particle at x is $|\Psi(x,t)|^2$. Ψ obeys the Schrödinger equation (SE):

$$i\hbar\Psi_t = -\frac{\hbar^2}{2m}\Psi_{xx} \quad \text{(free particle)}$$

where $\hbar$ is Planck's constant
m is particle mass

If the 'energy cost' of being at position x is $V(x)$, then the SE is

$$i\hbar\Psi_t = -\frac{\hbar^2}{2m}\Psi_{xx} + V(x)\Psi$$

Remarks:

- The particle is typically more likely to be at places where $V(x)$ is lower.

- Probability is conserved

$$\int_{-\infty}^{\infty} |\Psi|^2 dx = 1 \ \forall t$$

- Looking for solutions of the form

$$\Psi(x,t) = e^{-iEt/\hbar} \cdot \phi(x)$$

gives the following time=independent SE:

$$E\phi = -\frac{\hbar^2}{2m}\phi'' + V(x)\phi$$

Where E has the interpretation of 'energy'

For a particle in a '1d box,' $0 < x < l$, we have

$$E\phi = -\frac{\hbar^2}{2m}\phi'',\ 0 < x < l$$

with the following BCs

$$\phi(0) = \phi(l) = 0$$
$$\Rightarrow \text{standing wave solutions}$$

2.3 Lecture 6 - Higher Dimensional PDEs

Recap:

- 1st order PDEs, characteristics, change of coords.
- 2nd order PDEs (1 space dimension), diffusion/heat eq, with advection, decay, etc.

$$u_t = Du_{xx}$$

- Steady state solutions
- Wave equation

$$u_{tt} = c^2 u_{xx}$$

- Standing waves, with BCs and ICs.
- Dirichlet (u is known on boundary), and Neumann (u_x is known on boundary).
- ODEs: separation of variables, homogeneous solutions, 2nd order constant coefficients.

Higher dimensional PDEs (Logan 1.7)

- Most physical processes occur in more than 1d.
- We can generalize 1d derivations using vector calculus.

Heat flow in 2d/3d. 2d and 3d are similar, so we work in 3d. Consider heat flow in domain $\Omega \subseteq \mathbb{R}^3$.

- Let $u(x, y, z, t)$ be the temperature at point $(x, y, z) \in \Omega$ at time t.
- Let Cu be the thermal energy density.
- Let $\Phi(x, y, z, t)$ be the energy flux vector, i.e. Φ is the vector field describing heat flow.
- The rate of heat flow across a surface element, dS is $\Phi \cdot dS$
- Recall that $dS = n \cdot dA$, where n is the normal vector, and dA is the area element.
- In 2d, flow across a line element is $\Phi \cdot ndl$
- Let $f(x, y, z, t)$ be the rate at which heat is produced.
- Consider thermal energy in subdomain $B \subseteq \Omega$

Conservation/balance:

$$\frac{d}{dt}\int_B CudV = -\int_{\partial B} \Phi \cdot dS + \int_B fdV$$

$$\begin{array}{c}\text{change in}\\ \text{energy}\end{array} = \begin{array}{c}\text{flux out}\\ \partial B \text{ is boundary}\end{array} + \text{sources/sinks}$$

Remark: this applies to virtually every spatially varying density.

Using the divergence theorem, we can get the PDE.
Let $\vec{F}$ be a (smooth) vector field in region $B \subseteq \mathbb{R}^3$. Then

$$\int_{\partial B} F \cdot dS = \int_B \nabla \cdot F dB$$

where $\vec{F} = (F_1, F_2, F_3)$ and $\nabla \cdot F = \partial_x F_1 + \partial_y F_2 + \partial_z F_3$

Apply this to conservation equation:

$$\begin{aligned} \frac{d}{dt} \int_B Cu dV &= - \int_{\partial B} \Phi \cdot dS + \int_B f dV \\ \int_B Cu_t dV &= - \int_B \nabla \cdot \Phi dV + \int_B f dV \end{aligned}$$

This is true for any $B \subseteq \Omega$

$$\Rightarrow Cu_t = -\nabla \cdot \Phi + f$$

To get a PDE for u, we need a relationship between Φ and u. In 1 space dimension, we had $u = u(x,t)$, $\phi(x,t)$, and $\phi = -Ku_x$ with constant $K > 0$ (in 1d, ϕ is effectively a scalar).

Fick's Law in higher dimensions:

$$\begin{aligned} \Phi &= -K \nabla u \quad (\text{constant } K > 0) \\ \nabla u &= (u_x, u_y, u_z) \end{aligned}$$

This gives the PDE

$$\begin{aligned} Cu_t &= -\nabla \cdot \Phi + f \\ &= K \nabla \cdot (\nabla u) \end{aligned}$$

where

$$\begin{aligned} \nabla \cdot (\nabla u) &= \nabla \cdot (u_x, u_y, u_z) \\ &= u_{xx} + u_{yy} + u_{zz} \\ &:= \nabla^2 u \quad (\text{Laplacian of } u) \end{aligned}$$

This gives the PDE

$$\begin{aligned} Cu_t &= K\nabla^2 u + f \\ \text{for } &(x, y, z) \in \Omega \end{aligned}$$

Definition: The Laplacian
Let u be a scalar function which depends on space and possibly time. The Laplacian of u, $\nabla^2 u$ is the sum of 2nd derivatives with respect to space (in Cartesian coords).

Remarks:

- The Laplacian ∇^2 is a linear differential operator.
- We've only defined ∇^2 for scalar functions.
- Laplacian of u is sometimes written as Δu

$$\text{e.g. } \Delta u = \nabla^2 u = \nabla \cdot (\nabla u)$$

Heat/diffusion eq in higher dimensions.

$$u_t = D\nabla^2 u, \text{ constant } D > 0$$

for $u = u(x, y, z, t)$ and $(x, y, z) \in \Omega \subseteq \mathbb{R}^3$

Heat/diffusion eq with heat source.

$$u_t = D\nabla^2 u + f$$

for $f = f(x, y, z, t)$

Remarks:

- The time-independent cases also give important equations
  Poisson equation:

$$\begin{aligned} \nabla^2 u &= -f \quad \in \Omega \\ u &= u(x, y, z) \end{aligned}$$

Laplace equation:

$$\nabla^2 u = 0 \quad \in \Omega$$
$$u = u(x, y, z)$$

Wave equation:

$$u_{tt} = c^2 \nabla^2 u \quad \in \Omega$$

- These equations are fundamental for describing spatially varying quantities in 2d, 3d. E.g. fluid flow, vibrations, quantum mechanics, electromagnetism.

Boundary Conditions. Just like in 1d, PDEs in higher dimensions need BCs. Consider the heat equation $u_t = D\nabla^2 u$ in 1d, where $\Omega = [0, L] \subseteq \mathbb{R}$

- Dirichlet BCs: $\begin{cases} u(0,t) = g_1(t) \\ u(L,t) = g_2(t) \end{cases}$ i.e. u is known on boundary.
- Neumann BCs: $\begin{cases} u_x(0,t) = g_1(t) \\ u_x(L,t) = g_2(t) \end{cases}$ i.e. u_x is known on boundary (flux).

In 2d, we have the area $\Omega \in \mathbb{R}^2$ and $\partial\Omega$ is the boundary (curve).

- Dirichlet BC: $u(x, y, t) = g(x, y, t)$ for $(x, y) \in \partial\Omega$ (boundary).
  This can be written as
  $$u\big|_{\partial\Omega} = g$$
  i.e. u is known on boundary.
- Neumann BC: $n(x, y) \cdot \nabla u(x, y) = g(x, y, t)$ for $(x, y) \in \partial\Omega$ where n is the unit normal to $\partial\Omega$ at (x, y).
  This can be written as
  $$\vec{n} \cdot \nabla u\big|_{\partial\Omega} = g$$
  i.e. flux out of domain $n \cdot \nabla u$ is known.

$\boxed{\text{ex}}$ Vibrating drum.
A drum occupies region $\Omega \subseteq \mathbb{R}^2$, $u(x, y, t)$ is height of the drum, and the drum is clamped on its boundary.

$$u\big|_{\partial\Omega} = 0 \quad \text{(Dirichlet BC)}$$

The height of the drum obeys

$$\begin{aligned} u_{tt} &= c^2 \nabla^2 u \ \text{ on } \Omega \\ u\big|_{\partial\Omega} &= 0 \end{aligned}$$

where $c^2 = \frac{T}{\rho} = \frac{\text{tension}}{\text{density}}$

Standing wave solutions:

$$\nabla^2 u = -\frac{\omega^2}{c^2} u, \quad u\big|_{\partial\Omega} = 0$$

3 Unbounded Domains

3.1 Lecture 7 - Classification & Unbounded Heat Equation

Recap:

- Characteristics: 1st order PDEs
- Conservation laws and derivations for PDEs.
- Special solutions: steady states and standing waves.
- The Laplacian, ∇^2, which acts on spatial variables, e.g. for $u = u(x, y)$ or $u(x, y, t)$, we have $\nabla^2 u = u_{xx} + u_{yy}$.

Consider Laplace's equation for $u = u(x, y)$

$$\nabla^2 u = u_{xx} + u_{yy} = 0$$

Functions which satisfy $\nabla^2 u = 0$ are called harmonic, e.g. $x + y$, xy, $x^2 - y^2$.

- Laplace's equation comes with a domain $\Omega \subseteq \mathbb{R}^2$

$$\nabla^2 u = 0, \text{ for } (x, y) \in \Omega$$

  Note that $\Omega = \mathbb{R}^2$ is a possibility

- Laplace's equation also comes with BCs:
  - Dirichlet BCs: Specify u on $\partial\Omega$
  - Neumann BCs: Specify $n \cdot \nabla u$ on $\partial\Omega$

  where $\partial\Omega$ is the boundary of Ω and $\vec{n}$ is the outward unit normal to $\partial\Omega$

Classification of PDEs. We have 3 main PDEs for functions of 2 variables.

1) Diffusion: $u = u(x, t)$

$$u_t - ku_{xx} = 0, \quad k > 0$$

2) Wave: $u = u(x.t)$

$$u_{tt} - c^2 u_{xx} = 0$$

3) Laplace: $u = u(x, y)$

$$u_{xx} + u_{yy} = 0$$

Remarks:

- 1) and 2) are evolution equations which specify how u changes in time. Typically, prescribe an initial state, along with BCs, ask for state at time t.
- We think of 3) as a static problem.
- Choosing t to be a 'time' variable in 1) and 2), but not in 3), is not as arbitrary as it seems.

Classification:
Let $u(x, y)$ be a function of 2 variables. Here we think x, y, as general variables (not necessarily space or time). Consider a 2nd order PDE of the form:

$$Au_{xx} + Bu_{xy} + Cu_{yy} + F(x, y, u, u_x, u_y) = 0$$

where A, B, C are constants and F is some arbitrary function.

The discriminant of this PDE is

$$\boxed{D = B^2 - 4AC}$$

A PDE is called $\begin{cases} \text{hyperbolic if } D > 0 \\ \text{parabolic if } D = 0 \\ \text{elliptic if } D < 0 \end{cases}$

E.g.:

1) Heat equation $u_t - ku_{xx} = 0$, $k > 0$. Setting $y = t$ and comparing, we get $A = -k,\ B = 0,\ C = 0 \Rightarrow D = B^2 - 4AC = 0$. This means the heat equation is parabolic.

2) Wave equation $u_{tt} - c^2 u_{xx} = 0$. Setting $y = t$ gives $A = -c^2,\ B = 0,\ C = 1 \Rightarrow D = B^2 - 4AC = 4c^2 > 0$. This means the wave equation is hyperbolic.

3) Laplace's equation $u_{xx} + u_{yy} = 0$. This gives $A = 1,\ B = 0,\ C = 1 \Rightarrow D = B^2 - 4AC = -4 < 0$. This means Laplace's equation is elliptic.

Remarks:

- PDEs of the same class have similar behavior, e.g. hyperbolic equations are wave-time.
- Terminology comes from conic sections.
- In general, the coefficients A, B, C can be functions of (x, y), in which case, the PDE class could vary over the plane.

Having set up our 3 main classes of PDEs, we will proceed to solve them.

Unbounded Domains.

- Infinite or semi-infinite domains are easier because BCs are looser.
- We start with the heat equation.

Cauchy Problem: (Logan 2.1)
Consider the heat equation for $u(x,t)$ on the real line:

$$\begin{aligned} u_t = ku_{xx},\ k &> 0 \\ t &> 0 \\ x &\in \mathbb{R} \end{aligned}$$

We have IC $u(x,0) = \phi(x)$ with $x \in \mathbb{R}$. What about BCs?

- For infinite domains, we typically also have some decay conditions. Take $u_x \to 0$ as $x \to \pm\infty$.
- This is a pure IC problem, or a Cauchy problem.

Given a solution to the heat equation, we can generate more solutions. E.g. Suppose $G(x,t)$ solves the heat equation.

$$\begin{aligned} G_t - kG_{xx} &= 0 \\ \text{or } (\partial_t - k\partial_x^2)G &= 0 \end{aligned}$$

Then G_x also solves the heat equation:

$$\begin{aligned} (\partial_t - k\partial_x^2)G_x &= G_{xt} - kG_{xxx} \\ &= (G_t - kG_{xx})_x \\ &= 0 \end{aligned}$$

So if G solves the heat equation then G_x also solves the heat equation. It turns out that $G(x-y,t)$ also solves the heat equation.

$$(\partial_t - k\partial_x^2)G(x-y,t) = 0$$

and or any function $\psi(y)$,

$$\int_{-\infty}^{+\infty} G(x-y,t)\psi(y)dy$$

solves the heat equation.

$\boxed{\text{ex}}$ $w(x,t) = 2kt + x^2$ solves $w_t - kw_{xx} = 0$. Consider w_x as a solution.

$$w_x = 2x$$

This also solves the heat equation. Now consider $w(x-y,t) = 2kt - (x-y)^2$

$$\begin{aligned} \partial_t w(x-y,t) &= 2k \\ \partial_x^2 w(x-y,t) &= 2 \end{aligned}$$

What about $\int w(x-y,t)\psi(y)dy$? This will be in HW3.

Returning to the Cauchy problem, we have

$$\begin{aligned} u_t - ku_{xx} &= 0 \\ k,t > 0,\ x &\in \mathbb{R} \\ u(x,0) &= \phi(x) \\ u_x \to 0,\ \text{as } x &\to \pm\infty \end{aligned}$$

<u>Strategy</u>: First solve special case where ϕ is a step-function then use above moves to generate general solutions for any ϕ.

For step-function ϕ we have:

$$\begin{aligned} u_t - ku_{xx} &= 0 \\ k,t > 0,\ x &\in \mathbb{R} \\ u(x,0) &= \begin{cases} \alpha & x > 0 \\ 0 & x < 0 \end{cases} \end{aligned}$$

<u>Dimensional analysis</u>: The variables, constants here are: x, t, u, k, α. We can interpret

$$\begin{aligned} [u] &= \text{temp} \quad [x] = \text{length} \\ [\alpha] &= \text{temp} \quad [t] = \text{time} \end{aligned}$$

We also have

$$[u_t] = \frac{\text{temp}}{\text{time}} \quad [u_{xx}] = \frac{\text{temp}}{\text{length}^2}$$
$$\text{and } [k] = [\frac{u_t}{u_{xx}}] = \frac{\text{length}^2}{\text{time}}$$

There are two dimensionless quantities:

$$\begin{aligned} &\frac{u}{\alpha} \text{ from temperature parameters} \\ &\frac{x}{\sqrt{4kt}} \text{ from length, time} \\ &\Rightarrow \frac{u(x,t)}{\alpha} = f\left(\frac{x}{\sqrt{4kt}}\right) \end{aligned}$$

3.2 Lecture 8 - Cauchy Problem

Recap:

- Cauchy problem for heat equation (Logan 2.1)
- Goal: solve for $u(x,t)$

$$\begin{aligned} u_t - ku_{xx} = 0.\ k > 0 \\ x \in \mathbb{R},\ t > 0 \\ \text{IC:}\ \ u(x,0) = \phi(x) \end{aligned} \tag{$*$}$$

- Decay conditions, e.g. $u, u_x \to 0$ as $x \to \pm\infty$
  Here u is bounded, therefore $\phi(x)$ is bounded.
- The solution will look like $\phi(x)$ 'spreading out' over time.

How to solve the Cauchy problem?

1. First find solution when $u(x,0)$ is a step-function.
2. Take linear combinations & translates of this solution to solve the general Cauchy problem $(*)$.

So for the Cauchy problem above, we have

1. Step-function ICs

$$\begin{aligned} u_t - ku_{xx} = 0 \\ x \in \mathbb{R}, t > 0 \\ u(x,0) = \begin{cases} \alpha & x > 0 \\ 0 & x < 0 \end{cases} \end{aligned}$$

Use dimensional analysis as with last lecture, we continue with

$$\frac{u(x,t)}{\alpha} = f\left(\frac{x}{\sqrt{4kt}}\right)$$

Remarks:

- Any choice of dimensions for u, x, t would give us this equation above.
- Having obtained this expression, we 'forget' about dimensions and set $\alpha = 1$.

$$\begin{aligned} \Rightarrow u(x,t) &= f\left(\frac{x}{\sqrt{4kt}}\right) \\ &= f(z) \\ \text{for } z(x,t) &= \frac{x}{\sqrt{4kt}} \end{aligned}$$

- Trick: substitute $u(x,t) = f(z)$ into heat eq. and hope that we get an ODE for f
- We don't know if this will work, the above argument is slippery, full explanation involves 'symmetry.'

Now we transform the heat PDE into an ODE, recall that we have

$$\begin{aligned} u_t - ku_{xx} &= 0 \\ u(x,0) &= \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases} \end{aligned}$$

We look for solutions of the form $u(x,t) = f(z(x,t))$ where $z(x,t) = \frac{x}{\sqrt{4kt}}$.

We compute that

$$\begin{aligned}
u_t &= f'(z)\frac{\partial z}{\partial t} \\
&= -\frac{1}{2}\cdot\frac{x}{\sqrt{4kt^3}}f'(z) \\
&= \frac{-z}{2t}f'(z) \\
u_x &= f'(z)\frac{\partial z}{\partial x} \\
&= \frac{1}{\sqrt{4kt}}f'(z) \\
u_{xx} &= \frac{1}{4kt}f''(z)
\end{aligned}$$

So in all we have

$$\begin{aligned}
\frac{-z}{2t}f'(z) - k\frac{1}{4kt}f''(z) &= 0 \\
\frac{-z}{2t}f'(z) - \frac{1}{4t}f''(z) &= 0 \\
f'' + 2zf' &= 0
\end{aligned}$$

We've reduced the PDE to an ODE for $f(z)$. If we got an expression which involved other functions of x, t, we would NOT have an ODE.

The solutions of the ODE above is

$$\begin{aligned}
f'' + 2zf' &= 0 \\
\text{Set } h(z) &= f'(z) \\
\frac{dh}{dz} &= -2zh \\
\int \frac{1}{h}dh &= \int -2z dz \\
\ln h &= -z^2 + c \\
h(z) &= c_1 e^{-z^2} \\
f'(z) &= c_1 e^{-z^2} \\
f(z) &= c_1 \int_0^z e^{-s^2} ds + c_2
\end{aligned}$$

We plug $z = \frac{x}{\sqrt{4kt}}$ into the expression above

$$u(x,t) = \int_0^{\frac{x}{\sqrt{4kt}}} e^{-s^2} ds + c_2 \qquad (**)$$

Now we use the IC $u(x,0) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}$ Fix $x > 0$ and send $t \to 0$ in $(**)$ so that $\frac{x}{\sqrt{4kt}} \to \infty$. Now we have the following

$$1 = u(x,0) = c_1 \int_0^{\infty} e^{-s^2} ds + c_2$$

Now fix $x < 0$ and send $t \to 0$ so that $\frac{x}{\sqrt{4kt}} \to -\infty$. We have the following

$$0 = u(x,0) = c_1 \int_0^{-\infty} e^{-s^2} ds + c_2$$

The key result here is

$$\int_0^{\infty} e^{-s^2} ds = \int_{-\infty}^{0} e^{-s^2} ds = \frac{\sqrt{\pi}}{2} = \frac{1}{2} \int_{-\infty}^{\infty} e^{-s^2} ds$$

This means our IC becomes

$$\begin{aligned} c_1 \frac{\sqrt{\pi}}{2} + c_2 &= 1 \\ -c_1 \frac{\sqrt{\pi}}{2} + c_2 &= 0 \\ c_2 &= \frac{1}{2} \\ c_1 = \frac{1}{\sqrt{\pi}} & \end{aligned}$$

Definition. The error function $\mathrm{erf}(z)$ is defined by

$$\begin{aligned} \mathrm{erf}(z) &= \frac{2}{\sqrt{\pi}} \int_0^z e^{-s^2} ds \\ \Rightarrow \mathrm{erf}(0) &= 0 \\ \Rightarrow \lim_{z \to \infty} \mathrm{erf}(z) &= 1 \\ \Rightarrow u(x,t) &= \frac{1}{2}\left(1 + \mathrm{erf}\left(\frac{x}{\sqrt{4kt}}\right)\right) \end{aligned}$$

This gives us the solution to the heat equation $u_t - ku_{xx} = 0$ for $u(x,0) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}$

2. Now that we solved it for the special case of $u(x,0)$ being a step function, how do we solve the general Cauchy problem with $u(x,0) = \phi(x)$?

   Lemma. Suppose $v(x,t)$ solves the heat equation. Then the following are also solutions.

   - $v(x-y,t)$ for any y.
   - $a_1 v(x-y_1,t) + a_2 v(x-y_2,t)$ for any a_1, a_2, y_1, y_2.
   - $\int_a^b v(x-y,t)\psi(y)dy$ for any a, b, function ψ.
   - $v_x(x,t)$

   Let $F(x,t)$ be the solution with the step function IC.

$$\Rightarrow F(x,t) = \frac{1}{2}(1 + \mathrm{erf}(\frac{x}{\sqrt{4kt}}))$$

   Goal: use above moves to transform F into a solution of $(*)$. We will start small and go with a translation of the step function.

$$F(x-\epsilon,t)$$

$$u(x,0) = \begin{cases} \alpha & x > \epsilon \\ 0 & x < \epsilon \end{cases}$$

   Next, consider the difference

$$F(x+\frac{\epsilon}{2},t) - F(x-\frac{\epsilon}{2},t)$$

   This gives a rectangle, with width ϵ, height 1, centered at 0. Next, we try

$$\sum_{n=-\infty}^{\infty} \phi(n\epsilon)\left[F(x-n\epsilon+\frac{\epsilon}{2},t) - F(x-n\epsilon-\frac{\epsilon}{2},t)\right]$$

This gives a series of rectangles, with width ϵ, height $\phi(n\epsilon)$. Think Riemann sums. This is so cool! The solution is

$$\begin{aligned}\sum_{n=-\infty}^{\infty} \phi(n\epsilon)\left[F(x-n\epsilon+\frac{\epsilon}{2},t)-F(x-n\epsilon-\frac{\epsilon}{2},t)\right] \\ \approx \epsilon F_x(x-n\epsilon,t) \\ \to \int_{-\infty}^{\infty} F_x(x-y,t)\psi(y)dy\end{aligned}$$

3.3 Lecture 9 - Arbitrary Initial Conditions

Recap:

- Heat equation with step-function IC

$$\begin{aligned} u_t - ku_{xx} &= 0 \\ x \in \mathbb{R},\ t &> 0 \\ u(x,0) &= \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}\end{aligned}$$

- For solns, try $u(x,t) = f(z)$ for $z(x,t) = \frac{x}{\sqrt{4kt}}$
  $\Rightarrow f'' + 2zf' = 0$

- Soln

$$F(x,t) = \frac{1}{2}(1 + \text{erf}(\frac{x}{\sqrt{4kt}}))$$

  where

$$\begin{aligned}\text{erf}(z) &= \frac{2}{\sqrt{\pi}}\int_0^z e^{-s^2}ds \\ \text{erf}(0) &= 0 \\ \lim_{z\to\pm\infty} \text{erf}(z) &= \pm 1\end{aligned}$$

- If $v(x,t)$ is a soln to heat eq, then $v_x(x,t),\ v(x-y,t),\ \int v(x-y,t)\psi(y)dy$ are also solns to the heat equation.

Goal:

$$\begin{aligned} \text{Solve} \;\; u_t - ku_{xx} &= 0 \\ x \in \mathbb{R},\; t &> 0 \\ u(x,0) = \phi(x) \quad & \text{(bounded)} \end{aligned}$$

Strategy:

$$\begin{gathered} \text{Step-function IC} \\ \downarrow \\ \text{'Point source' IC} \\ \downarrow \\ \text{General IC} \end{gathered}$$

So the following are solutions

$$\begin{gathered} F(x,t) \\ \downarrow \\ F(x - \tfrac{\epsilon}{2}, t) \\ \downarrow \\ F(x + \tfrac{\epsilon}{2}, t) - F(x - \tfrac{\epsilon}{2}, t) \\ \downarrow \\ \epsilon F_x(x,t) \end{gathered}$$

where $F_x(x,t)$ is a solution for a 'point source.'

The IC ϕ can be thought of as a set of point sources of strength $\phi(y)$ at each $y \in \mathbb{R}$.

$$\int_{-\infty}^{\infty} \phi(y) F_x(x-y,t)dy$$

Remark: The point source function is also called the Dirac delta function

$$\delta(x) = \begin{cases} 0 & x \neq 0 \\ 1 & x = 0 \end{cases} \quad \text{s.t.} \quad \int_{-\infty}^{\infty} \delta(x)dx = 1$$

Let $G(x,t) = F_x(x,t)$

$$\Rightarrow G(x,t) = \partial_x \left[\frac{1}{2} + \frac{1}{2}\text{erf}\left(\frac{x}{\sqrt{4kt}} \right) \right]$$

Recall that $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-s^2} ds$

$$\Rightarrow G(x,t) = \frac{1}{\sqrt{4\pi kt}} \exp\left(\frac{-x^2}{4kt}\right)$$

G here is called the heat kernel or the fundamental solution to the heat equation.

The solution to the Cauchy problem

$$\begin{aligned} u_t - ku_{xx} &= 0 \\ x \in \mathbb{R},\ t &> 0 \\ u(x,0) &= \phi(x) \text{ (bounded)} \end{aligned}$$

is

$$\begin{aligned} u(x,t) &= \int_{-\infty}^{\infty} G(x-y,t)\phi(y)dy \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-y)^2}{4kt}} \cdot \phi(y)dy \end{aligned}$$

We can simplify this by substituting

$$\begin{aligned} s &= \frac{x-y}{\sqrt{4kt}} \\ ds &= \frac{-dy}{\sqrt{4kt}} \end{aligned}$$

gives the equivalent formula

$$u(x,t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-s^2} \phi(x - s\sqrt{4kt})ds$$

Remark:

- The above expression is called the Poisson integral representation.
- The solution is an integral, often with no closed form.

$\boxed{\text{ex}}$ Solve the following and give the solution in terms of erf.

$$u_t - ku_{xx} = 0$$
$$x \in \mathbb{R},\ t > 0$$
$$\text{with } u(x,0) = \phi(x) = \begin{cases} e^{\lambda x} & x < 0 \\ 0 & x > 0 \end{cases}$$

Using the Poisson integral above, we get

$$u(x,t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-s^2} \phi(x - s\sqrt{4kt}) ds$$
$$= \frac{1}{\sqrt{\pi}} \int_{\frac{x}{\sqrt{4kt}}}^{\infty} e^{-s^2} e^{\lambda x} e^{-\lambda s\sqrt{4kt}} ds$$

The limits are values of s for which $\phi(x - s\sqrt{4kt})$ is $\neq 0$

$$\Rightarrow x - s\sqrt{4kt} < 0 \Rightarrow u(x,t) = \frac{e^{\lambda x}}{\sqrt{\pi}} \int_{\frac{x}{\sqrt{4kt}}}^{\infty} e^{-(s^2 + 2\lambda s\sqrt{kt})} ds$$

Complete the square

$$s^2 + 2s\lambda\sqrt{kt} = (s + \lambda\sqrt{kt})^2 - \lambda^2 kt$$

Then we have

$$u(x,t) = \frac{e^{\lambda x}}{\sqrt{\pi}} \int_{\frac{x}{\sqrt{4kt}}}^{\infty} e^{-(s+\lambda\sqrt{kt})^2} e^{\lambda^2 kt}$$

Substitute $\theta = s + \lambda\sqrt{kt}$, $d\theta = ds$

$$\Rightarrow u(x,t) = \frac{e^{\lambda^2 kt + \lambda x}}{\sqrt{\pi}} \int_{\frac{x}{\sqrt{4kt}}}^{\infty} e^{-\theta^2} d\theta$$

Note that

$$\text{erf}(b) = \frac{2}{\sqrt{\pi}} \int_0^b e^{-s^2} ds$$
$$\frac{2}{\sqrt{\pi}} \int_0^b e^{-s^2} ds + \frac{2}{\sqrt{\pi}} \int_b^{\infty} e^{-s^2} ds = 1$$
$$\text{erf}(b) + \frac{2}{\sqrt{\pi}} \int_b^{\infty} e^{-s^2} ds = 1$$
$$\Rightarrow \frac{2}{\pi} \int_b^{\infty} e^{-s^2} ds = 1 - \text{erf}(b)$$

Therefore

$$u(x,t) = \frac{1}{2}e^{\lambda^2 kt + \lambda x}\left(1 - \operatorname{erf}\left(\frac{x}{\sqrt{4kt}} + \lambda\sqrt{kt}\right)\right)$$

$\boxed{\text{ex}}$ Solve the following and give the solution in terms of erf.

$$\begin{aligned} u_t - ku_{xx} &= 0 \\ x \in \mathbb{R},\ t &> 0 \\ u(x,0) = \phi(x) &= \begin{cases} x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

The solution is

$$u(x,t) = \frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty} e^{-s^2}\phi(x - s\sqrt{4kt})ds$$

$$\begin{aligned} &\phi \neq 0 \text{ between} \\ x - s\sqrt{4kt} = 0 &\Rightarrow s = \frac{x}{\sqrt{4kt}} = b \\ x - s\sqrt{4kt} = 1 &\Rightarrow s = \frac{x-1}{\sqrt{4kt}} = a \end{aligned}$$

$$\begin{aligned} \Rightarrow u &= \frac{1}{\sqrt{\pi}}\int_a^b e^{-s^2}(x - s\sqrt{4kt})ds \\ &= \frac{1}{\sqrt{\pi}}\left[x\int_a^b e^{-s^2}ds - \sqrt{4kt}\int_a^b se^{-s^2}ds\right] \\ &= \frac{1}{\sqrt{\pi}}\left[x \cdot \frac{\sqrt{\pi}}{2}(\operatorname{erf}(b) - \operatorname{erf}(a)) - \sqrt{4kt}[\frac{-1}{2}e^{-s^2}]_a^b\right] \\ &= \frac{x}{2}(\operatorname{erf}(b) - \operatorname{erf}(a)) + \sqrt{\frac{kt}{\pi}}(e^{-b^2} - e^{-a^2}) \end{aligned}$$

for $a = \dfrac{x-1}{\sqrt{4kt}},\ b = \dfrac{x}{\sqrt{4kt}}$

<u>Remarks</u>:

- The solutions here have infinite propagation speed, which is unphysical. Even if $u(x,0)=0$, we can have $u(x,t)\neq 0$, for small t, and $x\to\infty$.
- These solutions conserve mass. Suppose $u(x,t)$ is integrable and let

$$M(t)=\int_{-\infty}^{\infty}u(x,t)dx$$

then

$$\begin{aligned}\frac{dM}{dt}&=\int_{-\infty}^{\infty}u_t dx\\&=k\int_{-\infty}^{\infty}u_{xx}dx\\&=k\left[u_x\right]_{-\infty}^{\infty}\\\Rightarrow\frac{dM}{dt}&=0\text{ if }u_x\to 0\text{ as }x\to\pm\infty\\\Rightarrow M&\text{ is constant.}\end{aligned}$$

3.4 Lecture 10 - Unbounded Wave Equation

Recap:

- Heat equation on $\mathbb{R}$

$$\begin{aligned}u_t-ku_{xx}&=0\\k>0,\ t&>0\\x&\in\mathbb{R}\\u(x,0)&=\phi(x)\\&\text{(bounded)}\end{aligned}$$

- Solution to the heat equation with arbitrary initial condition.

$$u(x,t)=\int_{-\infty}^{+\infty}\frac{1}{\sqrt{4\pi kt}}e^{-\frac{(x-y)^2}{4kt}}\phi(y)dy$$

- Solution describes $u(x,t)$ 'spreading out' or diffusing

- If $u_x \to 0$ as $|x| \to \infty$ then total mass

$$M(t) = \int_{-\infty}^{\infty} u(x,t)dx$$

is conserved ($\frac{dM}{dt} = 0$)

Cauchy problem for wave equation. (Logan 2.2)
The wave equation for $u(x,t)$ is

$$\begin{aligned} u_{tt} - c^2 u_{xx} &= 0 \\ c &> 0 \end{aligned}$$

This has one space dimension, or 1d.

We can find the general solution more directly than for the heat equation.
<u>Method 1</u>:

$$\begin{aligned} \text{Let } \xi &= x - ct \\ \eta &= x + ct \end{aligned}$$

This is the same transformation as in HW3, where we write $u(x(\xi,\eta), t(\xi,\eta))$ and find $u_{\xi\xi}$, etc. Here we write $u(\xi(x,t), \eta(x,t))$

$$\begin{aligned} \Rightarrow u_x &= u_\xi \xi_x + u_\eta \eta_x \\ &= u_\xi + u_\eta \\ \Rightarrow u_{xx} &= \partial_\xi(u_\xi + u_\eta) \cdot \xi_x + \partial_\eta(u_\xi + u_\eta) \cdot \eta_x \\ &= u_{xx} + 2u_{\xi\eta} + u_{\eta\eta} \end{aligned}$$

and

$$\begin{aligned} u_t &= c^2 u_{\xi\xi} - 2c^2 u_{\xi\eta} + c^2 u_{\eta\eta} \\ \Rightarrow u_{tt} - c^2 u_{xx} &= -4c^2 u_{\xi\eta} \end{aligned}$$

so

$$\begin{aligned} u_{tt} - c^2 u_{xx} &= 0 \\ \Leftrightarrow u_{\xi\eta} &= 0 \end{aligned}$$

We now solve $u_{\xi\eta=0}$

$$u_\xi = f(\xi)$$
$$\Rightarrow u(\xi,\eta) = \int f(\xi)d\xi + G(\eta)$$

Notice that an integral of an arbitrary function is another arbitrary function

$$\Rightarrow u(\xi,\eta) = F(\xi) + G(\eta)$$
$$\Rightarrow u(x,t) = F(x-ct) + G(x+ct)$$

for arbitrary functions $F,\ G$

Method 2:

$$\text{Let } L = \partial_t^2 - c^2\partial_x^2$$

So $Lu = (\partial_t^2 - c^2\partial_x^2)u = u_{tt} - c^2u_{xx}$. We can factorize this differential operator.

$$\text{Let } L_1 = \partial_t - c\partial_x$$
$$L_2 = \partial_t + c\partial_x$$

Lemma: $L = L_1L_2$. Note that L_1L_2u means apply L_2, then apply L_1.

$$\begin{aligned} L_1L_2 &= L_1[(\partial_t + c\partial_x)u] \\ &= L_1[u_t + cu_x] \\ &= (\partial_t - c\partial_x)[u_t + cu_x] \\ &= u_{tt} + cu_{xt} - cu_{tx} - c^2u_{xx} \\ &= u_{tt} - c^2u_{xx} \\ &= Lu \end{aligned}$$

And so $Lu = 0 \Leftrightarrow L_1L_2u = 0$. The coordinates ξ,η from method 1 comes characteristics of 1st order PDEs

$$L_1v = 0$$
$$L_2v = 0$$

D'Alembert's solution of the wave equation. The Cauchy problem/initial value problem for wave equation is

$$
\begin{aligned}
u_{tt} - c^2 u_{xx} &= 0 \\
t &> 0 \\
x &\in \mathbb{R} \\
u(x,0) &= \phi(x) \\
u_t(x,0) &= \psi(x)
\end{aligned}
$$

e.g. $u(x,t)$ is the height of an infinite string with initial displacement $\phi(x)$ and initial velocity $\psi(x)$. The solution is

$$u(x,t) = F(x-ct) + G(x+ct)$$

We plug in the initial conditions to get

$$
\begin{aligned}
u(x,0) &= F(x) + G(x) = \phi(x) \\
u_t(x,0) &= -cF'(x) + cG'(x) = \psi(x)
\end{aligned}
$$

These can be solved for F, G and we get

$$u(x,t) = \frac{1}{2}(\phi(x-ct) + \phi(x+ct)) + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds$$

$\boxed{\text{ex}}$ Plucking an infinite string. Solve

$$
\begin{aligned}
u_{tt} - 4u_{xx} &= 0 \\
x \in \mathbb{R},\ t &> 0 \\
u(x,0) = \phi(x) &= e^{-x^2} \\
u_t(x,0) = \psi(x) &= 0
\end{aligned}
$$

So we have

$$
\begin{aligned}
u(x,t) &= \frac{1}{2}(\phi(x-ct) + \phi(x+ct)) \\
&= \frac{1}{2}(e^{-(x-2t)^2} + e^{-(x+2t)^2})
\end{aligned}
$$

Visually, notice that $e^{-(x-2t)^2}$ is a peak at location $x = 2t$, and as t increases, this peak moves to the right. In contrast, $e^{-(x+2t)^2}$ is a left-moving peak.

$\boxed{\text{ex}}$ $u_{tt} - c^2 u_{xx} = 0$

$$u(x,0) = 0$$
$$u_t(x,0) = \psi(x) = \frac{1}{1+x^2}$$

We use the D'Alembert solution again.

$$\begin{aligned} u(x,t) &= \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds \\ &= \frac{1}{2c}\int_{x-ct}^{x+ct} \frac{1}{1+s^2}ds \\ &= \frac{1}{2c}(\arctan(x+ct) - \arctan(x-ct)) \end{aligned}$$

Causality:
D'Alembert solution reveals a causal structure of the wave equation.

$$u(x,t) = \frac{1}{2}(\phi(x-ct) + \phi(x+ct)) + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds$$

Suppose $\phi(x),\ \psi = 0$ for $x < a,\ x > b$, i.e. ϕ, ψ are non-zero only when $x \in [a,b]$. Then $u(x,t)$ is non-zero only if $x - ct \in [a,b]$ or $x + ct \in [a,b]$.

- Signals propagate at speed $\pm c$ along lines

$$\begin{aligned} x - ct &= \text{ const} \\ x + ct &= \text{ const} \end{aligned}$$

- The area between $x + ct = a$ and $x - ct = b$ is called the region of influence of signal.
- The solution at a point (x_0, t_0) has a domain of dependence which comes from tracing characteristics backwards.

Energy. Just like the heat equation has a conserved mass, the wave equation has a conserved energy. Suppose $u_{tt} - c^2 u_{xx} = 0$ with u_t or $u_x \to 0$ as $|x| \to \infty$. We can define energy as

$$E(t) = \int_{-\infty}^{\infty} \frac{1}{2}(u_t^2 + c^2 u_x^2)dx$$

Then $\frac{dE}{dt} = 0$, i.e. E is conserved.

Remarks

- Conservation of $E(t)$ is a good check on solutions (especially for simulations)
- Conserved quantities tell us how solutions behaves.

Proof

$$\begin{aligned}
\frac{dE}{dt} &= \int_{\infty}^{\infty} (u_t u_{xx} + c^2 u_x u_{xt}) dx \\
&= \int_{-\infty}^{\infty} (c^2 u_t u_{xx} + c^2 u_x u_{xt}) dx \\
&= c^2 \int_{-\infty}^{\infty} \partial_x (u_t u_x) dx \text{ by product rule} \\
\Rightarrow \frac{dE}{dt} &= c^2 [u_t u_x]_{-\infty}^{\infty} \\
&= 0
\end{aligned}$$

Summary:

- Heat: parabolic, signal spreads out, diffuses
- Wave: hyperbolic, signal propagates at speed c, coherently
- Laplace: elliptic, ?

What about Laplace's equation?

$$u_{xx} + u_{yy} = 0 \text{ for } u(x, y)$$

Let's try to formulate a Cauchy/initial value problem as with heat/wave equations with y acting as time. So the Cauchy problem is

$$\begin{aligned}
u_{xx} + u_{yy} &= 0 \\
x \in \mathbb{R},\ y &> 0 \\
u(x, 0) &= \phi(x) \\
u_y(x, 0) &= \psi(x)
\end{aligned}$$

As with the wave equation, we have $u(x, 0)$ as the initial profile and $u_y(x, 0)$ as the initial velocity.

3.5 Lecture 11 - Heat Equation on Half-line

Recap:

- Heat equation Cauchy problem

$$\begin{aligned} u_t - ku_{xx} &= 0 \\ x &\in \mathbb{R} \\ k, t &> 0 \\ u(x,0) &= \phi(x) \end{aligned}$$

- Wave equation Cauchy problem

$$\begin{aligned} u_{tt} - c^2 u_{xx} &= 0 \\ x &\in \mathbb{R} \\ t &> 0 \\ u(x,0) &= \phi(x) \\ u_t(x,0) &= \psi(x) \end{aligned}$$

Following from last lecture, we have the following Cauchy problem for Laplace's equation.

$$\begin{aligned} u_{xx} + u_{yy} &= 0 \\ x \in \mathbb{R},\ y &> 0 \\ u(x,0) &= \phi(x) \\ u_y(x,0) &= \psi(x) \end{aligned}$$

where $u(x,0)$ is the initial profile and $u_y(x,0)$ is the initial velocity. This is fine for zero IC,

$$\begin{aligned} \text{Take}\quad \psi(x) &= 0 \\ \phi(x) &= \phi^0(x) = 0 \end{aligned}$$

Note here that superscripts are labels, not exponents. We have

$$u(x,y) = u^0(x,y) = 0$$

So zero IC gives zero solution, now we try 'almost zero' ICs.

$$\psi(x) = 0$$
$$\phi(x) = \phi^\lambda(x) = \lambda\cos(\frac{x}{\lambda})$$

Let $u^\lambda(x,y)$ be the solution for these ICs. We have $\lim_{\lambda\to 0}\phi^\lambda(x) = 0 = \phi^0(x)$. So does $\lim_{\lambda\to 0} u^\lambda(x,y) = 0$? No! But why?

$$u^\lambda(x,y) = \lambda\cos(\frac{x}{\lambda})\cosh(\frac{y}{\lambda})$$

This solution blows up as $\lambda \to 0$.

Remarks:

- The Cauchy problem (IVP) for Laplace's equation is ill-posed.
- Elliptic PDEs (like Laplace's equation) describe static, time-independent processes.
- A PDE with ICs or BCs is well-posed if solutions have
  - Existence
  - Uniqueness
  - Stability (solution depends continuously on ICs and BCs)
- The Cauchy problems for the heat and wave equation is well-posed.
- Next: The solutions to the wave/heat equation let us solve more general problems
  - Semi-infinite domains
  - Inhomogeneous problems

Semi-infinite domains (Logan 2.4). We've solved the wave/heat equations on domain $x \in \mathbb{R}$. We now solve PDEs on the half-line $x \in (0,\infty)$.
$\boxed{\text{ex}}$ Consider the heat equation on the half-line

$$u_t - ku_{xx} = 0$$
$$k, x, t > 0$$
$$u(x,0) = \phi(x) = xe^{-x^2}$$
$$u(0,t) = 0$$

Method: extend to an IVP on $\mathbb{R}$ using symmetry, whose solution on $x > 0$ is the desired solution.

Consider the heat equation on $\mathbb{R}$

$$\begin{aligned} v_t - kv_{xx} &= 0 \\ x \in \mathbb{R}, t &> 0 \\ u(x,0) = \begin{cases} \phi(x) & x > 0 \\ -\phi(-x) & x < 0 \end{cases} &= p(x) \end{aligned}$$

Here $p(x)$ is called the odd-extension of $\phi(X)$. The solution for $v(x,t)$ is

$$v(x,t) = \frac{1}{\sqrt{\pi}} \int e^{-s^2} p(x - s\sqrt{4kt}) ds$$

The solution $v(x,t)$ satisfies the half-line PDE for $x > 0$

$$\begin{aligned} \text{Check } \; & v_t - kv_{xx} = 0 \text{ for } x > 0 \\ & v(x,0) = p(x) = \phi(x) \text{ for } x > 0 \\ & v(0,t) = 0 \; \forall t \;\; \text{'by symmetry'} \end{aligned}$$

So the solution to the half-line PDE is

$$\begin{aligned} u(x,t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-s^2} p(x - s\sqrt{4kt}) ds \\ \text{for } x > 0 \end{aligned}$$

Remark: This works even for $\phi(0) \neq 0$

Odd-even functions. Let $f : \mathbb{R} \to \mathbb{R}$ be a function

- f is an even function if

$$f(x) = f(-x)$$

- f is an odd function if

$$f(x) = -f(-x)$$

Properties.

- If f is odd and continuous, then $f(0) = 0$.
- If f is even and smooth, then $f'(0) = 0$

Definition. Let $g : (0, \infty) \to \mathbb{R}$

- The odd extension of g is

$$G_{\text{odd}}(x) = \begin{cases} g(x) & x > 0 \\ -g(-x) & x < 0 \end{cases}$$

- The even extension of g is

$$G_{\text{even}}(x) = \begin{cases} g(x) & x > 0 \\ g(-x) & x < 0 \end{cases}$$

Symmetry of PDEs.
Suppose $u(x,t)$ is a solution of the heat or wave equation for $x \in \mathbb{R},\ t > 0$. If the initial data are odd functions, then $u(x,t)$ is odd wrt to x

$$\begin{aligned} u(x,t) &= -u(-x,t) \\ u(0,t) &= 0 \ \text{ if } u \text{ is continuous} \end{aligned}$$

Similarly, if the initial data are even functions, then $u(x,t)$ is even wrt to x

$$u_x(0,t) = 0\ \forall t \ \text{ if } u \text{ is smooth}$$

Solutions of half-line PDEs. $x \in (0, \infty)$.
Idea: Extend PDE to $x \in \mathbb{R}$ by taking an odd or even extension of initial data, depending on BCs.

$$\begin{matrix} \text{Dirichlet BCs} \\ u(0,t) = 0 \end{matrix} \rightarrow \begin{matrix} \text{odd extension} \\ \text{of ICs} \end{matrix} \rightarrow \begin{matrix} \text{solve} \\ \text{take } x > 0 \end{matrix}$$

$$\begin{matrix} \text{Neumann BCs} \\ u_x(0,t) = 0 \end{matrix} \rightarrow \begin{matrix} \text{even extension} \\ \text{of ICs} \end{matrix} \rightarrow \begin{matrix} \text{solve} \\ \text{take } x > 0 \end{matrix}$$

$\boxed{\text{ex}}$ Consider the wave equation.

$$\begin{aligned} u_{tt} - c^2 u_{xx} &= 0 \\ x > 0,\ t &> 0 \\ u(x,0) = \phi(x) &= \cos x, x > 0 \\ u_t(x,0) = \psi(x) &= 0, x > 0 \\ u_x(0,t) &= 0, t > 0 \end{aligned}$$

The BC here is Neumann, so we use even extension on the ICs. Let $p(x), q(x)$ be the even extensions of $\phi(x), \psi(x)$. Then

$$\begin{aligned} p(x) &= \begin{cases} \cos(x) & x > 0 \\ \cos(-x) & x < 0 \end{cases} \\ &= \cos x \quad \forall x \in \mathbb{R} \\ q(x) &= 0 \quad x \in \mathbb{R} \end{aligned}$$

We use D'Alembert's solution

$$\begin{aligned} u(x,t) &= \frac{1}{2}(p(x-ct) + p(x+ct)) \\ &= \frac{1}{2}(\cos(x-ct) + \cos(x+ct)) \\ &= \cos(x)\cos(ct), \quad x \in \mathbb{R} \end{aligned}$$

The half-line solution is

$$\begin{aligned} u(x,t) = \cos x \cos ct \\ x > 0 \end{aligned}$$

Our BC is satisfied

$$\begin{aligned} u_x &= -\sin x \cos ct \\ &= 0 \quad \text{for x} = 0 \end{aligned}$$

3.6 Lecture 12 - Inhomogeneous Wave/Heat Equation

Note on HW4, Q2a:

$$\begin{aligned} u_t - ku_{xx} = 0 \\ x \in \mathbb{R} \\ k > 0, t > 0 \\ u(x,0) = \exp\left(-\frac{(x-a)^2}{b^2}\right) \end{aligned}$$

Method 1: Poisson integral representation.
Method 2: Transform the heat kernel.

The heat kernel $G(x,t)$ satisfies heat equation

$$\begin{aligned} G(x,t) &= \frac{1}{\sqrt{4\pi kt}} e^{\frac{-x^2}{4kt}} \\ G_t - kG_{xx} &= 0 \end{aligned}$$

At time $t = 0$, $G(x,t)$ is a point source. For $t > 0$, $G(x,t)$ is Gaussian. Since $G(x,t)$ satisfies the heat equation, we have

- $G(x + A, t)$ satisfies heat equation
- $G(x + A, t + B)$ satisfies heat equation
- $\lambda G(x + A, t + B)$ satisfies heat equation

This problem has a Gaussian IC, and the heat kernel describes a sequence of Gaussians. To solve this problem, try the candidate solution

$$\begin{aligned} u(x,t) &= \lambda G(x + A, t + B) \\ &= \frac{\lambda}{\sqrt{4\pi k(t+B)}} \exp\left(-\frac{(x+A)^2}{4k(t+B)}\right) \end{aligned}$$

We need $u(x,0) = \exp\left(-\frac{(x-a)^2}{b^2}\right)$.

$$\begin{aligned} \Rightarrow A = -a,\ B = \frac{b^2}{4k},\ \lambda &= \sqrt{4\pi k B} \\ &= \sqrt{\pi} b \end{aligned}$$

Inhomogeneous PDEs
Consider the inhomogeneous heat equation with source $f(x,t)$

$$\begin{aligned} u_t - ku_{xx} &= f(x,t) \\ x \in \mathbb{R},\ t, k &> 0 \\ u(x,0) &= 0 \end{aligned}$$

<u>ODE Case</u>. Consider an ODE version of the above example, for $y(t)$

$$\begin{aligned} y' - ky &= t \\ t &> 0 \\ y(0) &= 0 \end{aligned}$$

Note here that the source is $f(t) = t$. We can solve this ODE in many ways, e.g. integrating factor.

$$\begin{aligned} e^{-kt}y' - ke^{-kt}y &= te^{-kt} \\ (e^{-kt}y)' &= te^{-kt} \\ e^{-kt}y &= \int_0^t e^{-k\tau}\tau d\tau + c \end{aligned}$$

Note that $y(0) = 0 \Rightarrow C = 0$

$$\begin{aligned} y(t) &= e^{kt}\int_0^t e^{-k\tau}\tau d\tau \\ &= \int_0^t e^{k(t-\tau)}\tau d\tau \end{aligned}$$

The solution at time t is the response to a source $f(\tau) = \tau$ for all $0 < \tau < t$. This is equivalent to taking solutions of the homogeneous ODE ($y' - ky = 0$) with IC $f(\tau) = \tau$ for $0 < \tau < t$ and superposing them.

E.g. define $w(t;\tau)$ by

$$\begin{aligned} w' - kw &= 0 \\ \text{with } w(0;\tau) &= \tau \end{aligned}$$

We solve for w

$$\Rightarrow w(t;\tau) = Ae^{kt} = \tau e^{kt}$$

The value $w(t-\tau;\tau)$ gives the effect of a source $f(\tau) = \tau$ at a time $t > \tau$.

Superpose to find $y(t)$

$$\begin{aligned} y(t) &= \int_0^t w(t-\tau;\tau)d\tau \\ &= \int_0^t e^{k(t-\tau)}\tau d\tau \end{aligned}$$

This is Duhamel's principle, which converts inhomogeneous problem with zero IC to homogeneous problem with non-zero IC.

PDE Case. Consider the inhomogeneous heat equation.

$$\begin{aligned} u_t - ku_{xx} &= f(x,t) \\ x \in \mathbb{R},\ k,t &> 0 \\ u(x,0) &= 0 \end{aligned}$$

First find solutions $w(x,t;\tau)$ of heat equation with IC $f(x,\tau)$

$$\begin{aligned} w_t - kw_{xx} &= 0 \\ x \in \mathbb{R},\ k,t &> 0 \\ w(x,0;\tau) &= f(x,\tau) \end{aligned}$$

Then solution is

$$u(x,t) = \int_0^t w(x,t-\tau;\tau)d\tau \qquad \text{(Duhamel's principle)}$$

We can write $w(x,t;\tau)$ in terms of the heat kernel $G(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{\frac{-x^2}{4kt}}$.

$$\begin{aligned} w(x,t;\tau) &= \int_{-\infty}^{\infty} G(x-y,t)f(y,\tau)dy \\ \Rightarrow u(x,t) &= \int_0^t d\tau \int_{-\infty}^{\infty} G(x-y,t-\tau)f(y,\tau)dy \end{aligned}$$

Remarks:

- Rarely possible to evaluate this integral.
- Expression is useful for analyzing solution properties, or numerics.
- Intuition: point heat source at position y, time τ, spreads out as $G(x-y,t-\tau)$
- Duhamel's solution due to source $f(x,t) =$ solution due to point sources of strength $f(y,\tau)$ at y,τ

4 Bounded Domains

4.1 Lecture 12 - Series Expansion

$\boxed{\text{ex}}$ Heat equation on a bounded domain in 1d.

$$
\begin{aligned}
u_t &= u_{xx} && \text{(I)} \\
0 < x &< \pi \\
t &> 0 \\
\text{BC: } u(0,t) &= u(\pi,t) = 0 && \text{(II)} \\
\text{IC: } u(x,0) &= f(x) && \text{(III)}
\end{aligned}
$$

Strategy:

1. Observe that

   $$u_n(x,t) = e^{-n^2 t}\sin(nx)$$

   Satisfies (I), (II) for $n = 1, 2, 3, \ldots$

2. Every solution to (I) and (II) is a superposition of u_n's

   $$u(x,t) = \sum_{n=1}^{\infty} a_n e^{-n^2 t}\sin(nx)$$
   $$\text{for } a_n \in \mathbb{R}$$

3. The IC determines the constants a_n

   $$u(x,0) = f(x) = \sum_{n=1}^{\infty} a_n \sin(nx)$$

   All suitable functions $g(x)$ on $x \in [0,\pi]$ (domain) can be uniquely written as a sum of $\sin(nx)$ terms.

Remarks:

- We will flesh this out in the phase of class.
- A key method is expanding functions in terms of 'simpler' functions (e.g. steps 2 & 3)

Series expansions.
Warm-up: Taylor series.
The Taylor series of a function $f(x)$ about $x = 0$ is

$$f(x) = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \dots$$

- The basis functions here are the functions in the expansion, i.e. $1, x, x^2, \dots$.
- The coefficients c_n can be calculated

$$c_n = \frac{f^{(n)}(0)}{n!}$$

- f is any sufficiently smooth functions defined at 0.

Fourier series. (Logan 3.2)
Let $f : [-\pi, \pi] \to \mathbb{R}$ (we will change domain of f later). The Fourier series of f is an expansion in terms of the following basis functions:

$$1,\ \cos(x),\ \cos(2x), \dots,\ \sin(x),\ \sin(2x), \dots$$

Fourier series for $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

The coefficients are given by

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx, \quad n = 0, 1, 2, \dots$$
$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx, \quad n = 1, 2, 3, \dots$$

4.2 Lecture 13 - Series Expansion cont.

Fourier series.
Let $f : [-\pi, \pi] \to \mathbb{R}$. The Fourier series for f is an expansion in terms of the following basis functions.

$$1 \text{ or } \cos(0 \cdot x),\ \cos x,\ \cos 2x,\ \dots,\ \sin x,\ \sin 2x$$

The Fourier series for $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

Remark:

- We can write Fourier series for more general domains (not just $[-\pi, \pi]$)
- We haven't yet said anything about when the series is valid.

The Fourier coefficients are given by

$$\begin{aligned} a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)dx, \quad n = 0, 1, 2, \dots \\ b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)dx, \quad n = 1, 2, 3, \dots \end{aligned}$$

Proof:
We will first check the expression for b_n

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)dx$$

We can calculate this using Fourier series for $f(x)$. Calculating $\int_{-\pi}^{\pi} f(x)\sin(nx)dx$ requires integrals of the form

$$\int_{-\pi}^{\pi}\cos(mx)\sin(nx)dx, \quad \int_{-\pi}^{\pi}\sin(mx)\sin(nx)dx$$
$$\forall m, n$$

First note that

$$\int_{-\pi}^{\pi}\sin(mx)dx = 0 \quad \forall m \in \mathbb{Z}$$

Check:

$$\begin{aligned} \int_{-\pi}^{\pi}\sin(mx)dx &= -\frac{1}{m}\cos(mx)\Big|_{-\pi}^{\pi} \\ &= -\frac{1}{m}[\cos(\pi m) - \cos(-\pi m)] \\ &= 0 \quad (\text{since } \cos(x) = \cos(-x)) \end{aligned}$$

Similarly

$$\int_{-\pi}^{\pi} \cos(mx)dx = \begin{cases} 0 & m \in \mathbb{Z},\ m > 0 \\ 2\pi & m = 0 \end{cases}$$

Also

$$\begin{aligned} \int_{-\pi}^{\pi} \cos(mx)\sin(nx)dx &= \int_{\pi}^{-\pi} \frac{1}{2}(\sin((n+m)x) - \sin((n-m)x))dx \\ &= 0 \end{aligned}$$

Similarly

$$\begin{aligned} \int_{-\pi}^{\pi} \sin(mx)\sin(nx)dx &= 0 \quad \text{for } m \neq n \\ &= \frac{1}{2}[\cos((m-n)x) - \cos((m+n)x)] \\ & m-n,\ m+n \neq 0,\ \text{if } m \neq n \end{aligned}$$

And

$$\int_{-\pi}^{\pi} \sin^2(nx)dx = \int_{-\pi}^{\pi} \cos^2(nx)dx = \pi$$

We use these integrals to calculate $\int_{-\pi}^{\pi} f(x)\sin(nx)dx$

$$\begin{aligned} &\int_{-\pi}^{\pi} f(x)\sin(nx)dx \\ &= \int_{-\pi}^{\pi} \left[\frac{a_0}{2} + \sum_{m=1}^{\infty}(a_m\cos(mx) + b_m\sin(mx))\right]\sin(nx)dx \\ &= \int_{-\pi}^{\pi} \frac{a_0}{2}\sin(nx)dx + \sum_{m=1}^{\infty}\left(\int_{-\pi}^{\pi} a_m\cos(mx)\sin(nx)dx + \int_{-\pi}^{\pi} b_m\sin(mx)\sin(nx)\right) \\ &= 0 + \sum_{m=1}^{\infty}\left(0 + \int_{-\pi}^{\pi} b_m\sin(mx)\sin(nx)\right) \\ &= \int_{-\pi}^{\pi} b_n\sin^2(nx)dx \\ &= b_n\pi \end{aligned}$$

<u>Remarks</u>:

- Key properties

$$\int_{-\pi}^{\pi} \cos(mx)\sin(nx)dx = 0$$

$$\int_{-\pi}^{\pi} \sin(nx)dx = 0$$

$$\int_{-\pi}^{\pi} \sin(mx)\sin(nx)dx = \begin{cases} 0 & m \neq n \\ \pi & m = n \end{cases}$$

- Functions whose product integrate to 0 are called <u>orthogonal</u>

$\boxed{\text{ex}}$ Fourier series of a step-function on $[-\pi, \pi]$

$$f(x) = \begin{cases} 1 & 0 < x < \pi \\ 0 & -\pi < x < 0 \end{cases}$$

Goal: find a_n, b_n such that

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos(nx) + b_n \sin(nx)) \\ a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)dx \quad n = 0, 1, 2, \ldots \\ b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)dx \quad n = 1, 2, 3, \ldots \end{aligned}$$

Calculate a_0

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{1}{\pi}\int_{0}^{\pi} 1dx = 1$$

For $n > 0$:

$$\begin{aligned} a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)dx \\ &= \frac{1}{\pi}\int_{0}^{\pi} \cos(nx)dx \\ &= \frac{1}{\pi}\left[\frac{1}{n}\sin(nx)\right]_0^{\pi} \\ &= 0 \end{aligned}$$

$$\begin{aligned}
b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)dx \\
&= \frac{1}{\pi}\int_{0}^{\pi} \sin(nx)dx \\
&= \frac{1}{\pi}\left[-\frac{1}{n}\cos(nx)\right]_0^{\pi} \\
&= \frac{1}{n\pi}(1-\cos(n\pi)) \\
&= \frac{1}{n\pi}(1-(-1)^n) \\
&= \begin{cases} 0 & n \text{ even} \\ \frac{2}{n\pi} & n \text{ odd} \end{cases}
\end{aligned}$$

$\Rightarrow$ Fourier series for $f(x)$ is

$$f(x) = \frac{1}{2} + \frac{2}{\pi}(\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \dots)$$

<ins>Remarks</ins>. As we take more terms, we get a better approximation for $f(x)$.

Function spaces. (Logan 3.2). Writing $f(x)$ in terms of Fourier basis is analogous to writing a vector in components.

$\boxed{\text{ex}}$ Consider $\mathbb{R}^3$. Let e_1, e_2, e_3 be the standard basis

$$\begin{aligned}
e_1 &= (1,0,0) \\
e_2 &= (0,1,0) \\
e_3 &= (0,0,1)
\end{aligned}$$

This basis is <ins>orthogonal</ins> since

$$e_1 \cdot e_2 = e_2 \cdot e_3 = e_1 \cdot e_3 = 0$$

This basis is <ins>normalized</ins> since

$$e_1 \cdot e_1 = e_2 \cdot e_2 = e_3 \cdot e_3 = 1$$

We can 'expand' vector $v \in \mathbb{R}^3$ in terms of e_1, e_2, e_3

$$\begin{gathered}
v = a_1 e_1 + a_2 e_2 + a_3 e_3 \\
\text{where } a_1 = v \cdot e_1, \text{ etc.}
\end{gathered}$$

Connection with Fourier series

vector $v, w \in \mathbb{R}^3$	$\rightarrow$	functions $f, g : [-\pi, \pi] \in \mathbb{R}$
inner product $v \cdot w$	$\rightarrow$	integral $\int_{-\pi}^{\pi} f(x)g(x)dx$
basis e_1, e_2, e_3	$\rightarrow$	Fourier basis $1, \cos x, \ldots, \sin x, \ldots$
expansion $v = \sum a_i e_i$ $a_i = v \cdot e_i$	$\rightarrow$	Fourier series $f(x) = \frac{a_0}{2} + \sum(a_n \cos nx + b_n \sin nx)$ $a_n = \frac{1}{\pi}(f, \cos nx),\ b_n = \frac{1}{\pi}(f, \sin nx)$

Notation: $(f, g) = \int_{-\pi}^{\pi} f(x)g(x)dx$

L^2-space. The correspondence above works because functions and vectors obey the same properties - in fact, we say that functions are vectors. To fix context, consider functions on a fixed domain $[a, b]$.

The function space $L^2[a, b]$ is the set of all functions $f : [a, b] \rightarrow \mathbb{R}$ such that $\int_a^b f(x)^2 dx < \infty$. This is called a square integrable function.

Fourier series, and other series expansions, work for L^2 functions. L^2 includes smooth functions but also some non-continuous functions like step-functions, etc. Typically, we only see L^2 functions. L^2 has the structure to understand series expansions.